

Constrained Reinforcement Learning for Unstable Point-Feet Bipedal Locomotion Applied to the Bolt Robot

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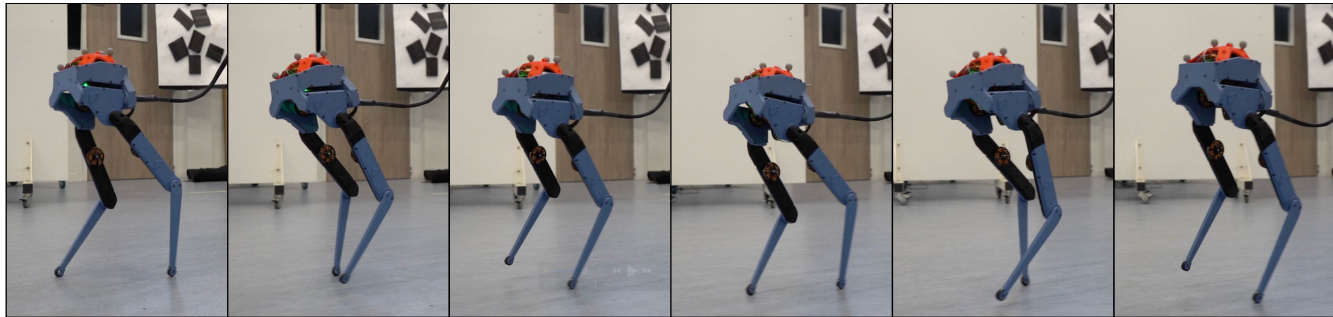


Fig. 1: The Bolt robot walking forward.

Abstract—Bipedal locomotion is a key challenge in robotics, particularly for robots like Bolt, which have a point-foot design. This study explores the control of such underactuated robots using constrained reinforcement learning, addressing their inherent instability, lack of arms, and limited foot actuation. We present a methodology that leverages Constraints-as-Terminations and domain randomization techniques to enable sim-to-real transfer. Through a series of qualitative and quantitative experiments, we evaluate our approach in terms of balance maintenance, velocity control, and responses to slip and push disturbances. Additionally, we analyze autonomy through metrics like the cost of transport and ground reaction force. Our method advances robust control strategies for point-foot bipedal robots, offering insights into broader locomotion. Videos and code are available at <https://gepetto.github.io/BoltLocomotion/>.

I. INTRODUCTION

Traditionally, Model Predictive Control (MPC) has been the dominant method for controlling legged robots [1], [2]. However, the emergence of advanced Reinforcement Learning (RL) algorithms [3] and the development of GPU-accelerated simulators [4], [5] have driven a significant shift toward learning-based policies [6], especially in the context of quadrupedal locomotion [7], [8]. In recent years, such techniques have also demonstrated great success in controlling bipedal and humanoid robots [9], [10], [11], enabling these robots to perform in increasingly complex environments and often outperforming traditional MPC-based approaches.

Despite these advancements, sim-to-real transfer remains challenging for robots with unconventional morphological traits, such as limited ground contact or underactuated limbs [12], [13], [14], [15], [16]. Among these, bipedal robots with

point feet [17], [18], [19], [20] pose a unique challenge due to their inherently unstable structure and minimal support area. While some prior works have achieved point-foot bipedal walking using quadrupedal robots [21], [22], these systems benefit from additional limbs that contribute to balance through angular momentum [23]—a strategy unavailable to arm-less bipedal robots. The scientific interest in point-foot bipedal robots is considerable.

As noted by Westervelt et al., “a model of walking with a point contact is an integral part of an overall model of walking that is more anthropomorphic in nature than the current flat-footed walking paradigm” [24]. Unlike rigid flat-footed robots—which often adopt overly simplified and non-anthropomorphic designs with stiff, unarticulated contact surfaces—point-foot systems better reflect the dynamics and control challenges seen in human locomotion. To explore this issue, we focus on the point-foot Bolt robot [25], a bipedal platform developed as part of the Open Dynamic Robot Initiative (ODRI). Bolt shares the same actuators as Solo, a quadrupedal robot from the same project, leveraging a modular and open-source design aimed at advancing torque-controlled legged locomotion. Notably, while Solo has been extensively studied [26], [27], [28], there has been relatively little research on Bolt [29], making it an interesting platform for our subject of interest.

In this paper, we address the control of bipedal point-foot robots without arms, using the Bolt platform as a representative system. We analyze its dynamics and identify key challenges that must be tackled to enable robust locomotion. Our contributions are as follows:

- **First Application of Constrained RL on Point-Foot Bipedal Locomotion:** We apply constrained reinforcement learning to the problem of point-foot bipedal locomotion, demonstrating its feasibility on a setting that has received limited attention in prior work.

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针对Bolt机器人的不稳定点足双足运动的约束强化学习

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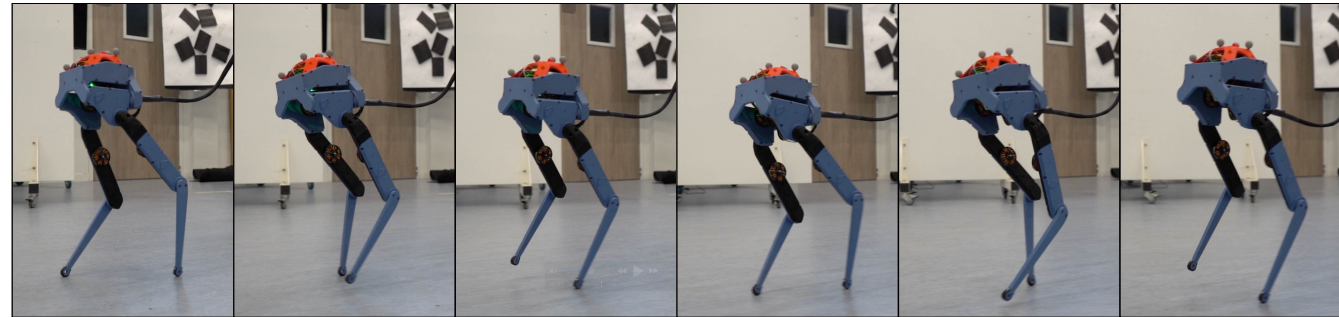


图1: Bolt机器人向前行走。

摘要——双足运动是机器人学中的一个关键挑战，尤其对于像Bolt这样采用点足设计的机器人而言。本研究探索了利用约束强化学习来控制此类欠驱动机器人，以应对其固有的不稳定性、缺乏手臂以及有限的足部驱动能力。我们提出了一种方法，该方法利用“约束即终止”和域随机化技术来实现仿真到现实迁移。通过一系列定性和定量实验，我们从平衡维持、速度控制以及对滑倒和推力扰动的响应等方面评估了我们的方法。此外，我们还通过运输成本和地面反作用力等指标分析了自主性。我们的方法推进了点足双足机器人的鲁棒控制策略，并为更广泛的运动研究提供了见解。视频和代码可在 <https://gepetto.github.io/BoltLocomotion/> 获取。

I. 引言

传统上，模型预测控制（MPC）一直是控制腿式机器人的主要方法 [1][2]。然而，先进强化学习（RL）算法的出现 [3]以及GPU加速模拟器的发展 [4], [5]，推动了向基于学习的策略的重大转变 [6]，尤其是在四足运动领域 [7], [8]。近年来，这些技术也在控制双足机器人和人形机器人方面取得了巨大成功 [9], [10], [11]，使这些机器人能够在日益复杂的环境中执行任务，并且通常优于传统的基于MPC的方法。

尽管取得了这些进展，但对于具有非常规形态特征（如有限的地面接触或欠驱动肢体）的机器人而言，仿真到现实迁移仍然具有挑战性 [12], [13], [14], [15], [16]。其中，双足机器人由于

点足 [17], [18], [19], [20] 结构本身的不稳定性和极小的支撑面积，构成了独特的挑战。虽然一些先前工作已利用四足机器人实现了点足双足行走 [21], [22]，但这些系统受益于额外的肢体，可通过角动量辅助平衡 [23]——这一策略对于无臂双足机器人而言并不可行。点足双足机器人的科学意义十分重大。

正如Westervelt等人所指出的，“具有点接触的行走模型是整体行走模型的一个组成部分，该模型在本质上比当前的平足行走范式更具拟人化特征” [24]。与刚性平足机器人（通常采用过度简化且非拟人化的设计，具有僵硬、无关节的接触面）不同，点足系统更能反映人类运动中的动力学和控制挑战。为了探讨这一问题，我们聚焦于点足Bolt机器人 [25]，这是一个作为开放动态机器人倡议（ODRI）一部分开发的双足平台。Bolt与同一项目中的四足机器人Solo共享相同的执行器，利用模块化和开源设计，旨在推进力矩控制的腿式运动。值得注意的是，虽然Solo已被广泛研究 [26], [27], [28]，但对Bolt的研究相对较少 [29]，这使其成为我们研究主题的一个有趣平台。

在本文中，我们以Bolt平台作为代表性系统，研究无手臂双足点足机器人的控制问题。我们分析了其动力学特性，并识别出实现稳健运动所必须应对的关键挑战。我们的贡献如下：

- **约束强化学习在点足双足运动中的首次应用：**我们将约束强化学习应用于点足双足运动问题，证明了其在先前工作中关注较少的场景下的可行性。

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- **Open-Source Training and Inference Pipeline:** We provide a fully open-source pipeline for training, inference, and logging on the low-cost, open-source Bolt robot, aiming to enhance reproducibility and support future benchmarking efforts.
- **Real-World Evaluation:** We conduct a real-world evaluation of the approach on the Bolt robot, providing insights into its performance and practical deployment.

The structure of the paper is as follows: Section II reviews related work on constrained RL and point-foot robots, Section III provides an in-depth description of the Bolt hardware, Section IV outlines the methodology, Section V presents experimental results, and Section VI concludes the paper.

II. RELATED WORK

Recent advances in GPU-accelerated simulators [4], [5] have enhanced the efficiency of RL for robotic locomotion by enabling massively parallel training, which reduces training times [7]. RL has been successful in quadrupedal locomotion, with policies trained in simulation effectively transferring to real-world robots [30], [8], [27], [21]. However, RL for bipedal locomotion remains more challenging due to the inherent instability of bipedal morphology [9], [10], [11], especially in point-foot bipedal robots, where the underactuated stance phase complicates the control [22]. The absence of arms further increases the challenge by eliminating mechanisms to counteract angular momentum [18], [31], [32]. The Bolt robot, a point-foot biped, embodies many of the fundamental challenges in dynamic legged locomotion. Previous work has leveraged MPC and whole-body MPC to achieve stable walking behaviors [33], [29], [34]. In this work, we instead focus on RL to develop more robust and adaptable locomotion policies that can better handle uncertainty and disturbances.

Transferring RL policies to real-world bipedal robots is hindered by issues like unmodeled dynamics, sensor noise, and hardware limitations [9], [10], [11]. Constrained RL frameworks enhance safety and robustness by incorporating constraints, such as joint position and torque limits, as termination conditions [26], [35], [28]. One notable approach, *CaT* (*Constraints as Terminations*), enforces these constraints during training to ensure hardware-compliant locomotion [26]. Alternative methods, such as [36], address similar challenges through different constraint-handling techniques. Furthermore, domain randomization techniques [12] expose policies to various training dynamics, such as variations in motor gains and friction coefficients, enhancing robustness to real-world discrepancies. Additionally, adaptive control strategies allow policies to adapt to real-world conditions by addressing discrepancies between simulation and reality [13], [14], [37]. In this work, we employ the *CaT* framework along with domain randomization to bridge the sim-to-real gap for the Bolt robot, allowing us to develop robust locomotion policies that account for real-world uncertainties.

Evaluation metrics for bipedal locomotion encompass multiple aspects of performance. Velocity tracking accuracy assesses how well the robot follows commanded velocity

profiles, ensuring precise motion control [38]. Maximum achievable speed evaluates the robot’s top velocity, reflecting its control limits [39], [40]. Autonomy is quantified through the cost of transport (CoT), which measures energy efficiency by calculating the energy consumed per unit distance traveled [41], [42]. Additionally, ground reaction force (GRF) profiles provide insights into balance, impact absorption, and adherence to physical constraints by analyzing force distribution during locomotion [43], [44]. Robustness is assessed through push recovery, which tests the robot’s ability to regain stability after external perturbations [45], [46], [47], and slippage recovery, which evaluates its ability to maintain balance when encountering unexpected loss of traction [38], [47]. In this work, we comprehensively evaluate the Bolt robot’s RL policies across all these metrics, analyzing its velocity tracking accuracy, maximum speed, energy efficiency, and robustness to external disturbances. This assessment offers insights into the behavior of the locomotion framework in real-world scenarios.

III. HARDWARE

A. Bolt Overview

Bolt (as shown in Fig. 1) is a bipedal robot developed by the ODRI [25]. It adopts a fully open-source, modular design philosophy aimed at enhancing accessibility and reproducibility in robotic research. Built with low-cost, off-the-shelf components and 3D-printed parts, Bolt features three torque-controlled degrees of freedom per leg and point feet. The robot omits arms entirely, which contributes to its lightweight form factor.

The Bolt robot relies on an external system, connected via a wired interface, for high-level computation and power supply. Commands are transmitted in real time via an Ethernet link to low-level motor controllers, enabling agile, closed-loop control while minimizing onboard complexity.

B. Locomotion Challenges

While Bolt’s streamlined design facilitates experimentation, it introduces several critical challenges for dynamic locomotion:

- **Limited Stability:** Point feet provide no flat contact surface, significantly reducing the support polygon and making balance control more demanding, particularly during dynamic motions or on uneven terrain [19].
- **Angular Momentum Regulation:** The absence of arms limits the robot’s ability to counterbalance body motion, placing a greater burden on the legs and torso to manage angular momentum during gait transitions and external disturbances [23].
- **Restricted Yaw Control:** Without an active yaw mechanism, Bolt struggles to reorient itself efficiently, which complicates tasks such as turning in confined spaces or navigating sharp trajectories [48].
- **Reduced Kinematic Redundancy:** Each leg’s three actuators limit the robot’s motion versatility, making agile behaviors such as running or adaptive stepping

- **开源训练与推理流水线:** 我们提供了一个完全开源的流水线，用于在低成本、开源的Bolt机器人上进行训练、推理和日志记录，旨在增强可复现性并支持未来的基准测试工作。

- **真实世界评估:** 我们在Bolt机器人上对该方法进行真实世界评估，深入分析了其性能与实际部署情况。

本文结构如下：第二部分回顾了约束强化学习与点足机器人的相关研究，第三部分详细描述了Bolt硬件，第四部分阐述了方法论，第五部分展示了实验结果，第六部分对全文进行总结。

II. 相关工作

近年来，GPU加速模拟器 [4], [5]通过实现大规模并行训练，提升了强化学习在机器人运动中的效率，从而缩短了训练时间 [7]。强化学习已在四足运动中取得成功，在仿真中训练的策略能够有效迁移至真实机器人 [30], [8], [27], [21]。然而，由于双足形态固有的不稳定性，强化学习在双足运动中仍面临更大挑战

[9], [10], [11],，尤其是在点足双足机器人中，欠驱动的支撑相使得控制更加复杂 [22]。手臂的缺失进一步增加了难度，因为消除了抵消角动量的机制 [18], [31][32]。Bolt机器人作为一种点足双足机器人，体现了动态腿式运动中的许多基本挑战。先前的研究利用模型预测控制和全身模型预测控制实现了稳定的行走行为

[33], [29], [34]。在本工作中，我们转而聚焦于强化学习，以开发更鲁棒、更具适应性的运动策略，从而更好地应对不确定性和干扰。

将强化学习策略迁移到真实世界双足机器人受到未建模动力学、传感器噪声和硬件限制等问题的阻碍

[9], [10], [11]。约束强化学习框架通过纳入约束（例如关节位置和力矩限制）作为终止条件来增强安全性和鲁棒性 [26], [35], [28]。一种值得注意的方法，*CaT*（约束作为终止），在训练过程中强制执行这些约束，以确保符合硬件限制的运动[26]。替代方法，如 [36]，，通过不同的约束处理技术应对类似挑战。此外，域随机化技术 [12] 使策略暴露于各种训练动力学中，例如电机增益和摩擦系数的变化，从而增强对现实差异的鲁棒性。另外，自适应控制策略允许策略通过解决仿真与现实之间的差异来适应现实世界条件 [13][14], [37]。在这项工作中，我们采用*CaT* 框架结合域随机化来弥合Bolt机器人的仿真到现实差距，从而开发出能够应对现实世界不确定性的鲁棒运动策略。

双足运动的评估指标涵盖多个性能方面。速度跟踪精度评估机器人遵循指令速度轮廓的程度，确保精确运动控制[v1]。

最大可达速度评估机器人的最高速度，反映其控制极限 [39], [40]。自主性通过运输成本（CoT）量化，该指标通过计算单位距离消耗的能量来衡量能效[41], [42]。此外，地面反作用力（GRF）轮廓通过分析运动过程中的力分布，为平衡、冲击吸收以及物理约束的遵守提供见解 [43], [44]。鲁棒性通过推扰恢复（测试机器人在外部扰动后恢复稳定的能力 [45], [46], [47].）和滑移恢复（评估机器人在遭遇意外牵引力丧失时维持平衡的能力 [38], [47]）进行评估。在本工作中，我们全面评估了Bolt机器人的强化学习策略在所有上述指标上的表现，分析了其速度跟踪精度、最大速度、能效以及对外部干扰的鲁棒性。这一评估为运动框架在真实场景中的行为提供了见解。

III. 硬件

A. Bolt概述

Bolt（如图1所示）是由ODRI [25]开发的一款双足机器人。它采用完全开源、模块化的设计理念，旨在提升机器人研究的可访问性和可复现性。Bolt采用低成本、现成的组件和3D打印部件制造，每条腿配备三个扭矩控制的自由度以及点足。机器人完全省略了手臂，这有助于实现其轻量化外形。

Bolt机器人依赖通过有线接口连接的外部系统进行高级计算和供电。指令通过以太网链路实时传输至低级电机控制器，从而实现敏捷的闭环控制，同时最大限度地降低机载复杂度。

B. 运动挑战

虽然Bolt的流线型设计便于实验，但它也为动态运动带来了几个关键挑战：

- **有限稳定性:** 点足缺乏平坦的接触面，显著缩小了支撑多边形，使得平衡控制更具挑战性，尤其是在动态动作或不平坦地形上 [19]。
- **角动量调节:** 缺少手臂限制了机器人平衡身体运动的能力，在步态转换和外部干扰期间，腿部和躯干调节角动量的负担随之加重 [23]。

- **受限偏航控制:** 由于缺乏主动偏航机制，Bolt难以高效地重新调整自身朝向，这使得在狭窄空间内转弯或沿急转弯轨迹行进等任务变得复杂 [48]。

- **降低的运动学冗余:** 每条腿的三个执行器限制了机器人的运动多样性，使得诸如跑步或自适应步态等敏捷行为

more complex to achieve and requiring sophisticated control strategies.

IV. METHOD

In order to train a locomotion policy for the point-foot robot, Bolt, we use a sim-to-real approach. We first train the policy in simulation with deep RL and then directly transfer it to the real robot. This section describes the main components used to support learning and sim-to-real transfer on the Bolt robot.

A. Constrained Reinforcement Learning

Consider a Constrained Markov Decision Process defined as $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \gamma, \mathcal{T}, \{\mathcal{C}^i\}_{i \in I})$, where $\mathcal{S}$ and $\mathcal{A}$ represent the state and action spaces, respectively, γ is the discount factor, $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^+$ is the reward function, and $\mathcal{T} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}^+$ defines the probabilistic transition dynamics. The system is subject to constraints $\{\mathcal{C}^i : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}\}_{i \in I}$, where each constraint $\mathcal{C}^i$ yields a scalar penalty signal $c^i \in \mathbb{R}^+$ for a given state-action pair (s, a) . We look for a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximizes the discounted sum of future rewards:

$$\max_{\pi} \mathbb{E}_{\tau \sim \pi, \mathcal{T}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right], \quad (1)$$

while ensuring the constraints satisfaction:

$$\mathbb{P}_{(s,a) \sim \rho_{\gamma}^{\pi, \mathcal{T}}} [c^i(s, a) > 0] \leq \epsilon_i, \quad \forall i \in I. \quad (2)$$

We solve the problem defined by (1) and (2) using the *CaT* framework [26], built on Proximal Policy Optimization (PPO) [3], which incorporates constraints as termination conditions during training.

B. States and Actions

The state vector $s \in \mathcal{S}$ comprises the commanded base velocity $(\mathbf{v}^{*\top} \ \omega^*)^{\top}$, where $\mathbf{v}^*$ and ω^* represents respectively the desired linear and angular velocities in the robot’s base frame. It also includes the base angular velocity ω , the projected gravity, the joint positions $\mathbf{q}$, the joint velocities $\dot{\mathbf{q}}$, and the previous actions. The six-dimensional action space corresponds to the angular position commands for the Bolt robot’s joints. The policy outputs scaled delta actions, which are added to a predefined reference angular configuration and then tracked by a high-frequency proportional-derivative (PD) controller.

C. Rewards

The reward function is designed to promote precise tracking of the commanded base velocity, with separate terms for tracking the linear velocity in the xy -plane and the yaw velocity. The reward of the linear velocity is defined as:

$$R_{xy}(s) = \exp \left(-\frac{\|\mathbf{v}^* - \mathbf{v}\|^2}{\sigma_1^2} \right) \quad (3)$$

where $\mathbf{v}$ denote the measured linear velocity expressed in the robot’s base frame, and σ_1 represents a scaling factor.

TABLE I: Constraints applied during training. A constraint is considered violated when its associated cost $c_i > 0$.

Safety Constraints	
Knee or base collision	$c_{\text{knee/base contact}} = \mathbb{1}_{\text{knee/base contact}}$
Upside-down state	$c_{\text{upsidedown}} = \mathbb{1}_{\text{upsidedown}}$
Foot contact force limit	$c_{\text{foot contact } j} = \ f^{\text{foot } j}\ _2 - f^{\text{lim}}$
Torque limits	$c_{\text{torque } k} = \tau_k - \tau^{\text{lim}}$
Joint velocity limits	$c_{\text{joint velocity } k} = \dot{q}_k - \dot{q}^{\text{lim}}$
Joint acceleration limits	$c_{\text{joint acceleration } k} = \ddot{q}_k - \ddot{q}^{\text{lim}}$
Posture Constraints	
Base orientation	$c_{\text{ori}} = \ \text{base ori}_{xy}\ _2 - \text{base}^{\text{lim}}$
Hip orientation	$c_{\text{hip } l} = \text{hip ori}_l - \text{hip}^{\text{lim}}$
Knee orientation	$c_{\text{knee } l} = \text{knee ori}_l - \text{knee}^{\text{lim}}$
Gait Constraints	
Walking (single foot contact)	$c_{\text{foot contact}} = n_{\text{foot contact}} - 1 $
Jumping (alternating zero or two foot contacts)	$c_{\text{foot contact}} = n_{\text{foot contact}} \bmod 2 $
Foot air time	$c_{\text{air time } j} = t_{\text{air time}}^{\text{des}} - t_{\text{air time } j}$

Similarly, the reward for the yaw velocity is defined as:

$$R_{\text{yaw}} = \exp \left(-\frac{(\omega^* - \omega)^2}{\sigma_2^2} \right) \quad (4)$$

where σ_2 represents a scaling factor.

D. Constraints

To ensure safe and transferable locomotion, we define three categories of constraints during training: **safety constraints**, **posture constraints**, and **gait constraints**. These are detailed in Table I.

a) Safety Constraints: These constraints prevent critical failures that could damage the robot. They include prohibiting knee or base collisions with the ground, avoiding upside-down states, and limiting excessive foot contact forces $f^{\text{foot } j}$ of foot j with a threshold f^{lim} . Actuator safety is ensured by limiting joint torques τ_k , joint velocities $\dot{q}_k$, and accelerations $\ddot{q}_k$, where k denotes the actuator index. These are constrained within limits τ^{lim} , $\dot{q}^{\text{lim}}$, and $\ddot{q}^{\text{lim}}$, respectively.

b) Posture Constraints: Posture constraints regulate the robot’s body configuration. The base orientation in the roll and pitch axes is constrained to prevent excessive tilting, with a limit defined by base^{lim} . Similarly, the orientations of the hip joints hip_l , and the knee joints knee_l (where l indicates the leg index), are bounded by limits hip^{lim} and knee^{lim} , ensuring a consistent posture during movement.

c) Gait Constraints: These constraints define the robot’s locomotion gait. The walking constraint ensures that only one foot is in contact with the ground at any given time, expressed as $|n_{\text{foot contact}} - 1|$, where $n_{\text{foot contact}}$ denotes the number of feet in contact with the ground. For jumping, the constraint enforces that either zero or two feet are in contact with the ground, represented by $|n_{\text{foot contact}} \bmod 2|$. Foot air-time constraints ensure that each foot stays in the air for a sufficient amount of time during each locomotion cycle, defined by $t_{\text{air time}}^{\text{des}} - t_{\text{air time } j}$, where $t_{\text{air time } j}$ is the actual time that foot j is in the air.

Switching between walking and jumping behaviors requires retraining the network due to the mutually exclusive nature of the constraints.

更难以实现，并需要复杂的控制策略。

IV. 方法

为了训练点足机器人Bolt的运动策略，我们采用了一种仿真到现实的方法。首先在仿真环境中通过深度强化学习训练策略，然后直接将其迁移到真实机器人上。本节将介绍支持Bolt机器人学习与仿真到现实迁移的主要组件。

A. 约束强化学习

考虑一个约束马尔可夫决策过程，定义为 $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \gamma, \mathcal{T}, \{\mathcal{C}^i\}_{i \in I})$ ，其中 $\mathcal{S}$ 和 $\mathcal{A}$ 分别表示状态空间和动作空间， γ 是折扣因子， $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^+$ 是奖励函数， $\mathcal{T} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}^+$ 定义了概率转移动力学。该系统受约束 $\{\mathcal{C}^i : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}\}_{i \in I}$ 的限制，其中每个约束 $\mathcal{C}^i$ 针对给定的状态-动作对 (s, a) 产生一个标量惩罚信号 $c^i \in \mathbb{R}^+$ 。我们寻找一个策略 $\pi : \mathcal{S} \rightarrow \mathcal{A}$ ，以最大化折扣后的未来奖励总和：

$$\max_{\pi} \mathbb{E}_{\tau \sim \pi, \mathcal{T}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right], \quad (1)$$

同时确保满足约束条件：

$$\mathbb{P}_{(s,a) \sim \rho_{\gamma}^{\pi, \mathcal{T}}} [c^i(s, a) > 0] \leq \epsilon_i, \quad \forall i \in I. \quad (2)$$

我们使用基于近端策略优化（PPO） [3], 的*CaT* 框架 [26], 求解由(1)和(2)定义的问题，该框架在训练过程中将约束作为终止条件纳入考量。

B. 状态与动作

状态向量 $s \in \mathcal{S}$ 包含指令基座速度 $(\mathbf{v}^{*\top} \ \omega^*)^{\top}$ ，其中 $\mathbf{v}^*$ 和 ω^* 分别表示机器人基座坐标系中期望的线速度和角速度。此外，状态向量还包括基座角速度 ω 、投影重力、关节位置 $\mathbf{q}$ 、关节速度 $\dot{\mathbf{q}}$ 以及先前的动作。六维动作空间对应Bolt机器人关节的角位置指令。策略输出经过缩放的动作增量，这些增量被添加到预定义的参考角配置中，随后由高频比例-微分（PD）控制器进行跟踪。

C. 奖励

奖励函数旨在促进对指令基座速度的精确跟踪，其中包含分别用于跟踪 xy 平面内的线速度和偏航速度的独立项。线速度的奖励定义如下：

$$R_{xy}(s) = \exp \left(-\frac{\|\mathbf{v}^* - \mathbf{v}\|^2}{\sigma_1^2} \right) \quad (3)$$

其中 $\mathbf{v}$ 表示在机器人基座坐标系中测量的线速度， σ_1 代表缩放因子。

表I: 训练过程中应用的约束。当相关成本 $c_i > 0$ 时，该约束被视为违反。

安全约束	
膝盖或基座碰撞 倒立状态	$c_{\text{knee/base 接触}} = \mathbb{1}_{\text{knee/base contact}}$
足部接触力限制	$c_{\text{upsidedown}} = \mathbb{1}_{\text{upsidedown}}$
力矩限制	$c_{\text{foot 接触 } j} = \ f^{\text{foot } j}\ _2 - f^{\text{lim}}$
关节速度限制	$c_{\text{torque } k} = \tau_k - \tau^{\text{lim}}$
关节加速度限制	$c_{\text{joint 速度 } k} = \dot{q}_k - \dot{q}^{\text{lim}}$
	$c_{\text{joint 加速度 } k} = \ddot{q}_k - \ddot{q}^{\text{lim}}$
姿态约束	
基座朝向	$c_{\text{ori}} = \ \text{基座朝向}_{xy}\ _2 - \text{基座}^{\text{lim}}$
髋关节方向	$c_{\text{hip } l} = \text{髋关节方向}_l - \text{髋关节}^{\text{lim}}$
膝关节方向	$c_{\text{knee } l} = \text{膝盖或}_l - \text{膝盖}^{\text{lim}}$
步态约束	
行走（单脚触地）	$c_{\text{foot contact}} = n_{\text{foot contact}} - 1 $
跳跃（交替零或双脚触地） 脚部腾空时间	$c_{\text{foot contact}} = n_{\text{foot contact}} \bmod 2 $ $c_{\text{air time } j} = t_{\text{air time}}^{\text{des}} - t_{\text{air time } j}$

类似地，偏航速度的奖励定义如下：

$$R_{\text{yaw}} = \exp \left(-\frac{(\omega^* - \omega)^2}{\sigma_2^2} \right) \quad (4)$$

其中 σ_2 表示一个缩放因子。

D. 约束

为了确保安全且可迁移的运动，我们在训练过程中定义了三种约束类别：**安全约束**、**姿态约束**和**步态约束**。这些内容详见表I。

*a) 安全约束：*这些约束用于防止可能损坏机器人的关键故障。它们包括禁止膝盖或基座与地面碰撞、避免倒置状态，以及限制足部接触力 $f^{\text{foot } j}$ （足部 j ）超过阈值 f^{lim} 。执行器安全通过限制关节扭矩 τ_k 、关节速度 $\dot{q}_k$ 和加速度 $\ddot{q}_k$ 来确保，其中 k 表示执行器索引。这些值分别被限制在 τ^{lim} 、 $\dot{q}^{\text{lim}}$ 和 $\ddot{q}^{\text{lim}}$ 的范围内。

*b) 姿态约束：*姿态约束用于调节机器人的身体构型。基座在横滚轴和俯仰轴上的朝向受到约束，以防止过度倾斜，其限制由基座 $^{\text{lim}}$ 定义。类似地，髋关节髋和膝关节膝 $_l$ （其中 l 表示腿部索引）的朝向分别被限制在髋 $^{\text{lim}}$ 和膝 $^{\text{lim}}$ 的范围内，从而确保运动过程中姿态的一致性。

*c) 步态约束：*这些约束定义了机器人的运动步态。行走约束确保在任何时刻只有一只脚与地面接触，表示为 $|n_{\text{foot 接触}} - 1|$ ，其中 $n_{\text{foot 接触}}$ 表示与地面接触的脚的数量。对于跳跃，约束强制要求零只或两只脚与地面接触，表示为 $|n_{\text{foot contact}} \bmod 2|$ 。足部腾空时间约束确保每个脚在每个运动周期内保持足够长的腾空时间，定义为 $t_{\text{air time}}^{\text{des}} - t_{\text{air time } j}$ ，其中 $t_{\text{air time } j}$ 是脚 j 实际腾空的时间。

由于约束的互斥性质，在行走与跳跃行为之间切换需要重新训练网络。

E. Domain Randomization

To ensure robust zero-shot sim-to-real transfer, we employ domain randomization during training, as summarized in Table II. We introduce variations in both the environment and the robot’s dynamics, training the policy to generalize better to real-world uncertainties. Additionally, noise is injected into the observations to simulate sensor imperfections and further improve robustness.

a) Environment and Dynamics Randomization: First, we randomize terrain properties by varying the ground friction coefficient and introducing uneven terrain profiles. The robot’s dynamics is also randomized by modifying joint friction, shifting the center of mass (CoM) of the base, and perturbing the masses and inertia tensors of the robot’s links. By randomizing the physics parameters, we prevent the policy from overfitting to both the simulator and the specific URDF model of the robot, ensuring it remains robust to real-world variations rather than exploiting simulator-specific artifacts.

b) State and Command Perturbations: To prevent overfitting to a fixed initial condition, the robot’s initial state is randomized by applying random offsets to joint positions and velocities at spawn. External forces are randomly applied to the base to simulate real-world disturbances, and additional latency effects are introduced by simulating actuator delays and command variations.

c) Observation Noise: To account for sensor imperfections, noise is added to the observations. This noise injection further enhances robustness by preventing the policy from relying on overly precise state estimates.

F. Key Factors for Sim-to-Real Transfer

Among all randomization parameters, joint friction variations and actuator delays are particularly critical for successful sim-to-real transfer. Without accurately modeling joint friction, the robot fails to maintain balance and collapses immediately upon deployment. Likewise, actuator delays introduce inherent latency that must be accounted for to ensure smooth and stable motion execution. While other randomization parameters enhance overall robustness, these two factors are found to be indispensable for achieving reliable real-world locomotion.

V. EXPERIMENTS

We conducted quantitative and qualitative experiments on the real robot to evaluate our controller, highlighting its strengths and identifying potential limitations.

A. Experimental Setup

The policies were trained using the IsaacLab framework [4], leveraging the PPO implementation from the CleanRL library [49], [3]. Our approach is implemented within our *CaT* framework [26], available on GitHub¹.

After successful training in simulation, the learned policies were directly deployed on the real Bolt bipedal robot,

TABLE II: Domain randomization parameters for training. Key sim-to-real factors are in bold.

Category	Parameter	Distribution
Terrain	Ground friction coefficient	$\mathcal{U}(0.4, 1.5)$
	Height noise	$\mathcal{U}(0.0, 0.02)$ m, step = 0.005 m
Robot Dynamics	Mass scale factor	$\mathcal{U}(0.8, 1.2)$
	Inertia scale factor	$\mathcal{U}(0.8, 1.2)$
	Base CoM displacement	$\mathcal{U}(-0.02, 0.02)$ m
	Joint friction coefficient	$\mathcal{U}(0.01, 0.1)$
Initial State	Base position (x, y)	$\mathcal{U}(-0.5, 0.5)$ m
	Base yaw angle	$\mathcal{U}(-\pi, \pi)$ rad
	Base linear velocity	$\mathcal{U}(-0.3, 0.3)$ m/s
	Base angular velocity	$\mathcal{U}(-0.1, 0.1)$ rad/s
	Joint pos./vel. scale factor	$\mathcal{U}(0.9, 1.1)$
Events	Actuation delay	$\mathcal{U}(0, 2)$ simulation steps
	Push linear velocity (x, y)	$\mathcal{U}(-0.5, 0.5)$ m/s
	Push linear velocity (z)	$\mathcal{U}(-0.1, 0.1)$ m/s
	Push angular velocity	$\mathcal{U}(-0.5, 0.5)$ rad/s
Observation Noise	Base angular velocity	$\mathcal{U}(-0.2, 0.2)$ rad/s
	Projected gravity noise	$\mathcal{N}(0, 0.05^2)$ with bias $\mathcal{U}(0, 0.05)$
	Joint position noise	$\mathcal{U}(-0.01, 0.01)$ rad with bias $\mathcal{U}(0, 0.05)$ rad
	Joint velocity noise	$\mathcal{U}(-1.5, 1.5)$ rad/s

described in Sec. III. The policy runs at a frequency of 50Hz on an Apple Mac M3 Max CPU. Target joint positions are transmitted to the onboard PD controller, operating at a frequency of 10kHz. The complete inference pipeline, including code and logs, is available on GitHub¹ for reproducibility.

B. Evaluation Metrics

To quantitatively evaluate the performance of our approach, we introduce distinct evaluation metrics across three axes:

- **Performance Axis:** These metrics evaluate the robot’s maximum velocity and the accuracy of its velocity control.
- **Autonomy Axis:** These metrics evaluate the short-term autonomy by monitoring power consumption, and long-term autonomy based on the ground reaction force computed for each foot.
- **Robustness Axis:** These metrics evaluate the controller’s capacity to reject external disturbances.

Additionally, we conducted slip recovery and jumping experiments to further demonstrate the robustness and adaptability of the controller.

1) Performance Metrics:

a) Velocity Accuracy: To assess the accuracy of the velocity tracking of our controller, we computed the steady-state error for a set of velocity references along either the x -axis and the y -axis. The velocity tracking error $\bar{\epsilon}$ is defined as the difference between the reference velocity $\mathbf{v}^*$ and the robot’s mean steady-state velocity $\bar{\mathbf{v}}$.

b) Maximum Velocity: The maximum velocity is defined as the highest steady-state mean velocity attained along the x or the y -axis, in both directions, as the velocity reference is progressively increased. The process continues until the robot ceases to accelerate.

2) Autonomy Metrics:

a) Short-Term Autonomy: To assess the short-term autonomy of our controller, we measure the CoT for different

E. 域随机化

为确保鲁棒的零样本仿真到现实迁移，我们在训练过程中采用域随机化，如表II所示。我们引入环境和机器人动力学两方面的变化，训练策略以更好地泛化到现实世界的不确定性。此外，在观测值中注入噪声以模拟传感器缺陷，从而进一步提升鲁棒性。

*a) 环境与动力学随机化：*首先，我们通过改变地面摩擦系数和引入不平坦地形轮廓来随机化地形属性。机器人的动力学也通过修改关节摩擦、移动基座质心以及扰动机器人连杆的质量和惯性张量来进行随机化。通过随机化物理参数，我们防止策略过度拟合模拟器和机器人的特定URDF模型，确保其对现实世界的变化保持鲁棒性，而不是利用模拟器特有的伪影。

*b) 状态与指令扰动：*为了防止过度拟合固定的初始条件，机器人的初始状态通过在生成时对关节位置和速度施加随机偏移来进行随机化。外力被随机施加到基座上以模拟现实世界的干扰，并通过模拟执行器延迟和指令变化引入额外的延迟效应。

*c) 观测噪声：*为了考虑传感器的不完美性，在观测中添加了噪声。这种噪声注入通过防止策略依赖过于精确的状态估计，进一步增强了鲁棒性。

F. 仿真到现实迁移的关键因素

在所有随机化参数中，关节摩擦变化和执行器延迟对于成功的仿真到现实迁移尤为关键。如果未能精确建模关节摩擦，机器人将无法维持平衡，并在部署后立即倒塌。同样，执行器延迟会引入固有延迟，必须加以考虑以确保平稳稳定的运动执行。虽然其他随机化参数能增强整体鲁棒性，但这两个因素被认为是实现可靠现实世界运动不可或缺的。

V. 实验

我们在真实机器人上进行了定量和定性实验，以评估我们的控制器，突出其优势并识别潜在局限性。

A. 实验设置

策略使用IsaacLab框架[4], 进行训练，并利用了CleanRL库 [49], [3]中的PPO实现。我们的方法在我们的*CaT* 框架 [26],中实现，该框架可在GitHub¹上获取。

在仿真中成功训练后，学习到的策略被直接部署到真实的Bolt双足机器人上，

¹<https://gepetto.github.io/BoltLocomotion/>

表II：训练的域随机化参数。Keysim到真实因子以粗体显示。

类别	参数	分布
地形	地面摩擦系数	$\mathcal{U}(0.4, 1.5)$
	高度噪声	$\mathcal{U}(0.0, 0.02)$ 米, 步长 = 0.005 米
机器人动力学	质量缩放因子	$\mathcal{U}(0.8, 1.2)$
	惯性缩放因子	$\mathcal{U}(0.8, 1.2)$
	基座质心位移	$\mathcal{U}(-0.02, 0.02)$ 米
	关节摩擦系数	$\mathcal{U}(0.01, 0.1)$
初始状态	基座位置 (x, y)	$\mathcal{U}(-0.5, 0.5)$ 米
	基座偏航角	$\mathcal{U}(-\pi, \pi)$ 弧度
	基座线速度	$\mathcal{U}(-0.3, 0.3)$ 米/秒
	基座角速度	$\mathcal{U}(-0.1, 0.1)$ 弧度/秒
	关节位置/速度比例因子	$\mathcal{U}(0.9, 1.1)$
事件	执行延迟	$\mathcal{U}(0, 2)$ 仿真步数
	推动线速度 (x, y)	$\mathcal{U}(-0.5, 0.5)$ 米/秒
	推动线速度 (z)	$\mathcal{U}(-0.1, 0.1)$ 米/秒
	推动角速度	$\mathcal{U}(-0.5, 0.5)$ 弧度/秒
观测噪声	基座角速度	$\mathcal{U}(-0.2, 0.2)$ 弧度/秒
	投影重力噪声	$\mathcal{N}(0, 0.05^2)$ 偏置 $\mathcal{U}(0, 0.05)$
	关节位置噪声	$\mathcal{U}(-0.01, 0.01)$ 弧度, 偏置 $\mathcal{U}(0, 0.05)$ 弧度
	关节速度噪声	$\mathcal{U}(-1.5, 1.5)$ 弧度/秒

如第三节所述。该策略在Apple Mac M3 Max CPU上以50Hz的频率运行。目标关节位置被传输至板载PD控制器，该控制器以10kHz的频率运行。完整的推理流程（包括代码和日志）已在GitHub¹ 上公开，以确保可复现性。

B. 评估指标

为了定量评估我们方法的性能，我们引入了三个轴上的不同评估指标：

• **性能轴：**这些指标评估机器人的最大速度及其速度控制的准确性。

• **自主性轴：**这些指标通过监测功耗来评估短期自主性，并基于每只脚计算的地面反作用力来评估长期自主性。

• **鲁棒性轴：**这些指标评估控制器抵抗外部干扰的能力。

此外，我们还进行了滑移恢复和跳跃实验，以进一步展示控制器的鲁棒性和适应性。

1) 性能指标:

*a) 速度精度：*为了评估我们控制器的速度跟踪精度，我们计算了一组沿 x 轴和 y 轴的速度参考值的稳态误差。速度跟踪误差 ϵ 定义为参考速度 $\mathbf{v}^*$ 与机器人的平均稳态速度 $\bar{\mathbf{v}}$ 之间的差值。

*b) 最大速度：*最大速度定义为：随着速度参考值逐步增加，沿 x 轴或 y 轴（两个方向）所能达到的最高稳态平均速度。该过程持续进行，直至机器人停止加速。

2) 自主性指标:

*a) 短期自主性：*为了评估控制器的短期自主性，我们测量了不同速度参考值下的运输成本，这些参考值沿轴和轴方向设定。

¹<https://gepetto.github.io/BoltLocomotion/>

velocity references along the x and y -axes. The CoT is defined as:

$$\text{CoT} = \frac{\sum_{i=0}^{N-1} \sum_{k=0}^{n_{\mathbf{u}}-1} P_{i,k}^{\text{loss}}}{Nmg \|\bar{\mathbf{v}}\|_2} \quad (5)$$

where N is the total number of time steps during the experiment, $n_{\mathbf{u}}$ is the number of joints, m is the robot's mass, g is the gravitational acceleration, and $\bar{\mathbf{v}}$ denotes the robot's average velocity throughout the experiment. The power loss $P_{i,k}^{\text{loss}}$ for actuator k at timestep i is:

$$P_{i,k}^{\text{loss}} = P_{i,k}^J + P_{i,k}^f \quad (6)$$

where $P_{i,k}^J$ and $P_{i,k}^f$ are the Joule and friction power losses, respectively, as described by Fadini et al. [50].

b) Long-Term Autonomy: To evaluate the potential damage caused by foot impacts on the ground, we estimate the ground reaction force using the following equation of motion:

$$M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau}(\mathbf{u}) + \mathbf{J}_c(\mathbf{q})^T \mathbf{f} \quad (7)$$

where $\mathbf{q}$ and $\dot{\mathbf{q}}$ represent the measured joint positions and velocities, respectively, $\boldsymbol{\tau}$ denotes the joint torques generated by the control inputs $\mathbf{u}$, $\mathbf{J}_c$ is the contact Jacobian of the feet, $\ddot{\mathbf{q}}$ is the joint acceleration, and $\mathbf{f}$ is the ground reaction force.

This equation is solved using well-established algorithms from the literature [51], [52]. The computed ground reaction forces are then analyzed as part of our benchmark to assess impact-related wear and long-term autonomy.

3) Robustness Criteria:

a) Push Recovery: To quantify the robot's ability to reject external disturbances, an instrumented stick is used to apply controlled pushes to the robot's base. The pose of the stick and the force exerted on the robot are measured and the applied force is progressively increased in different directions until the robot loses stability and falls. The maximum impulse sustained before falling is recorded. The impulse is computed as the discrete integral:

$$\lambda = \delta t \sum_{i=1}^{M-1} \frac{f_i + f_{i-1}}{2} \quad (8)$$

where M denotes the total number of time steps during the push event, f_i represents the measured force at time step i , and δt is the time step duration.

C. Results

1) Performance Criteria:

a) Velocity Accuracy: Velocity commands range from -1 m/s to 1 m/s along the x and y -axes separately, with increments of 0.1 m/s. Each experiment is repeated at least three times per velocity reference. The steady-state velocity is determined by applying a moving average filter and extracting the stable segment of the signal. Results are presented in Fig. 3, where all velocities are normalized by $\sqrt{gL}$ (with L being the robot's leg length), corresponding to the Froude number normalization, as outlined in [53].

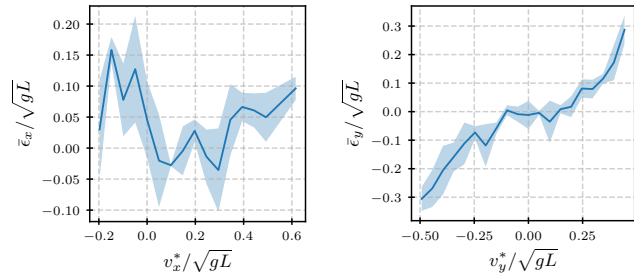


Fig. 3: Normalized steady-state velocity error versus normalized velocity command along the x -axis (left) and the y -axis (right). The solid line represents the mean, and the shaded area indicates the standard deviation.

TABLE III: Minimum and maximum normalized velocities achieved along the x and y axes.

	$v_{\min}/\sqrt{gL}$	$v_{\max}/\sqrt{gL}$
x -axis	-0.34	0.54
y -axis	-0.31	0.30

This normalization enables meaningful comparisons across different locomotion speeds and morphologies.

Overall, the robot successfully moves in the commanded direction (e.g., forward when instructed to do so). Along the x -axis, it achieves significantly higher forward velocities than backward ones but exhibits considerable drift, leading to poor velocity tracking. In contrast, along the y -axis, it demonstrates symmetric performance in both left and right directions. These results confirm that the robot is not only capable of balancing but also of following velocity commands. However, drift becomes more pronounced at higher velocity references. This drift can be attributed to the absence of linear velocity measurements in our setup, which eliminates direct feedback necessary for accurate reference tracking. As a result, the robot lacks immediate velocity feedback, making it harder to adjust movements in real-time for accurate control. This issue is compounded by the robot's inherently unstable morphology, further amplifying the challenge of maintaining precise control at higher speeds.

b) Maximum Velocity: Velocity references along the desired axis increase in increments of 0.1 m/s until the robot falls. The maximum steady-state velocity achieved is recorded. Table III summarizes the resulting performance, with all velocities normalized as before.

It is worth emphasizing that the robot achieves higher velocities in forward motion compared to backward, while exhibiting symmetric performance in lateral directions. When walking forward, the robot achieves a Froude number normalization of $Fr = 0.54$, which is comparable to the typical human walking value of $Fr \simeq 0.5$. Despite lower maximum velocities in the backward and lateral directions, the robot's performance remains competitive. Qualitative observations from video footage indicate that prior work [29] resulted in significantly lower velocities, highlighting the improved agility of our approach.

沿 x 轴和 y 轴的速度参考值。运输成本定义为:

$$\text{CoT} = \frac{\sum_{i=0}^{N-1} \sum_{k=0}^{n_{\mathbf{u}}-1} P_{i,k}^{\text{loss}}}{Nmg \|\bar{\mathbf{v}}\|_2} \quad (5)$$

其中 N 是实验期间的总时间步数, $n_{\mathbf{u}}$ 是关节数量, m 是机器人质量, g 是重力加速度, $\bar{\mathbf{v}}$ 表示机器人在整个实验中的平均速度。执行器 k 在时间步 i 的功率损耗 $P_{i,k}^{\text{loss}}$ 为:

$$P_{i,k}^{\text{loss}} = P_{i,k}^J + P_{i,k}^f \quad (6)$$

其中 $P_{i,k}^J$ 和 $P_{i,k}^f$ 分别是焦耳功率损耗和摩擦功率损耗, 如 Fadini 等人 [50]所述。

b) 长期自主性: 为了评估脚部撞击地面造成的潜在损伤, 我们使用以下运动方程估算地面反作用力:

$$M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau}(\mathbf{u}) + \mathbf{J}_c(\mathbf{q})^T \mathbf{f} \quad (7)$$

其中 $\mathbf{q}$ 和 $\dot{\mathbf{q}}$ 分别表示测量的关节位置和速度, $\boldsymbol{\tau}$ 表示由控制输入 $\mathbf{u}$ 产生的关节扭矩, $\mathbf{J}_c$ 是脚部的接触雅可比矩阵, $\ddot{\mathbf{q}}$ 是关节加速度, $\mathbf{f}$ 是地面反作用力。

该方程使用文献 [51], [52]中成熟的算法求解。计算出的地面反作用力随后作为我们基准测试的一部分进行分析, 以评估冲击相关磨损和长期自主性。

3) 鲁棒性准则:

a) 推扰恢复: 为了量化机器人抵抗外部干扰的能力, 使用一根仪器化杆对机器人基座施加受控推挤。测量杆的位姿以及施加在机器人上的力, 并逐步增加不同方向上的施加力, 直到机器人失去稳定性并摔倒。记录摔倒前承受的最大冲量。该冲量通过离散积分计算:

$$\lambda = \delta t \sum_{i=1}^{M-1} \frac{f_i + f_{i-1}}{2} \quad (8)$$

其中 M 表示推挤事件期间的总时间步数, f_i 表示在时间步 i 时测得的力, δt 是时间步长。

C. 结果

1) 性能准则:

a) 速度精度: 速度指令范围从 -1 米/秒到 1 米/秒, 分别沿 x 轴和 y 轴, 步长为 0.1 米/秒。每个实验针对每个速度参考值至少重复三次。稳态速度通过应用移动平均滤波器并提取信号的稳定段来确定。结果如图3所示, 其中所有速度均通过 $\sqrt{gL}$ 进行归一化 (L 为机器人的腿长), 对应于弗劳德数归一化, 如 [53]所述。

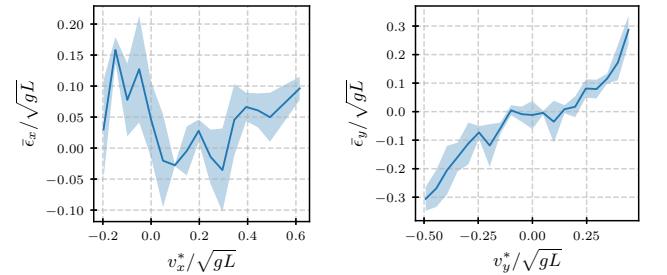


图3: 沿 x 轴 (左) 和 y 轴 (右) 的归一化稳态速度误差与归一化速度指令的关系。实线表示均值, 阴影区域表示标准差。

表III: 沿 x 轴和 y 轴达到的最小和最大归一化速度。

	$v_{\min}/\sqrt{gL}$	$v_{\max}/\sqrt{gL}$
x 轴	-0.34	0.54
y 轴	-0.31	0.30

这种归一化使得能够在不同的运动速度和形态之间进行有意义的比较。

总体而言, 机器人能够成功沿指令方向移动 (例如, 当被指示向前时向前移动)。沿 x 轴, 其前向速度显著高于后向速度, 但存在明显的漂移, 导致速度跟踪效果不佳。相比之下, 沿 y 轴, 机器人在左右两个方向上表现出对称的性能。这些结果证实, 机器人不仅能够保持平衡, 还能遵循速度指令。然而, 在较高的速度参考值下, 漂移变得更加明显。这种漂移可归因于我们的设置中缺乏线速度测量, 从而消除了精确参考跟踪所需的直接反馈。因此, 机器人缺少即时的速度反馈, 使得难以实时调整运动以实现精确控制。机器人本身固有的不稳定形态进一步加剧了这一问题, 从而放大了在更高速度下维持精确控制的挑战。

b) 最大速度: 沿目标轴的速度参考值以 0.1 米/秒的增量增加, 直至机器人摔倒。记录达到的最大稳态速度。表III总结了由此产生的性能, 所有速度均按之前的方式进行了归一化处理。

值得强调的是, 机器人在向前运动时能达到比向后运动更高的速度, 同时在横向方向上表现出对称的性能。当向前行走时, 机器人达到了弗劳德数归一化值 $Fr = 0.54$, 这与典型的人类行走值 $Fr \simeq 0.5$ 相当。尽管在向后和横向方向上的最大速度较低, 但机器人的性能仍然具有竞争力。来自视频录像的定性观察表明, 先前工作 [29] 导致的速度显著较低, 这凸显了我们方法在敏捷性上的提升。

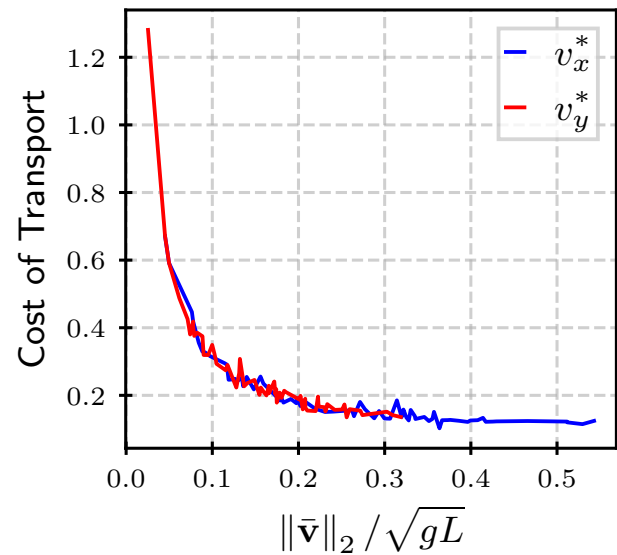


Fig. 4: Cost of transport as a function of the steady-state velocity norm. Blue represents velocity references along the x -axis, while red corresponds to those along the y -axis.

2) Autonomy Criteria:

a) Short-Term Autonomy: The CoT is evaluated at various steady-state velocities of the robot. Results are categorized based on the velocity command direction (forward/backward or left/right) and are presented in Fig. 4.

For velocity references along both the x and y -axes, the CoT decreases as the steady-state velocity norm increases. It can be used in future work to model battery consumption for an embedded version of the robot. Additionally, it establishes a useful benchmark for comparison in future studies.

b) Long-Term Autonomy: The mean ground reaction force (GRF) is found to be independent of the robot's velocity, remaining at $90\% \pm 6.2\%$ of the robot's weight. This value being below 100% is explained by the presence of short double support phases—periods during which both feet briefly share the load. Since the robot primarily walks with single-foot support, the average GRF per foot remains below full body weight. This indicates reduced GRF during gait transitions, which contributes to mechanical longevity. Furthermore, it confirms that the contact force constraint imposed during training is maintained in practice.

3) Robustness Criteria:

a) Push Recovery: The robot was subjected to over 300 pushes, covering a full range of directions around its base. Fig. 5 presents the maximum impulse, normalized as a percentage of the robot's weight, that the robot could withstand without falling, categorized in 45° quadrants.

Overall, the robot exhibits greater robustness to recover from pushes applied to the back and sides. Conversely, it is generally less robust when pushed from the front or at the corners, except for the upper right corner. This anomaly can be attributed to insufficient data points recorded in that region. For comparison, an impulse of 4% of the robot's

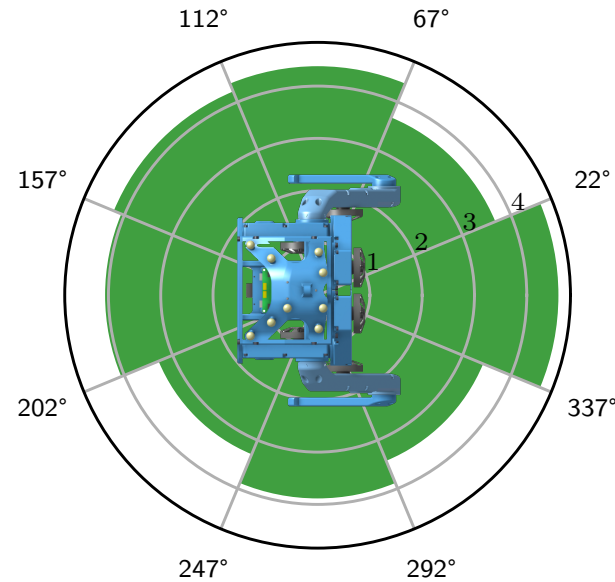


Fig. 5: Maximum impulse before falling (expressed as % of weight $\times$ seconds) as a function of the push angle, where 0° corresponds to a push applied to the back of the robot's base.

weight over one second is equivalent to a 70 kg human being struck by a 2.8 kg ball over the course of 1 second, demonstrating the robustness of our method.

D. Additional Results

To qualitatively demonstrate the robustness of the balancing process, we present additional results. Fig. 6 shows the robot recovering from a slip after the carpet was pulled out while it was walking on it. These images are extracted directly from the accompanying video, which provides a more dynamic visualization of the entire recovery sequence.

Fig. 7 shows the robot performing jumps, enabled by modifying the gait constraint to allow ballistic phases (see Sec.IV-D). It successfully completed over 50 consecutive jumps, maintaining this behavior for more than 15 seconds, as demonstrated in the accompanying video.

VI. CONCLUSION

This work explored the control of a point-foot bipedal robot, emphasizing its significance in achieving anthropomorphic locomotion. Unlike flat-footed bipeds, point-foot robots face unique stability challenges, making their control an interesting research topic.

We developed a constrained RL approach and successfully transferred policies from simulation to the real robot without additional fine-tuning. Our experiments demonstrated that the robot can balance, track velocity commands, jump, and resist external disturbances, showcasing a high level of robustness.

Despite these achievements, real-world deployment remains challenging due to hardware fragility and external factors, such as the use of a wire connecting the robot

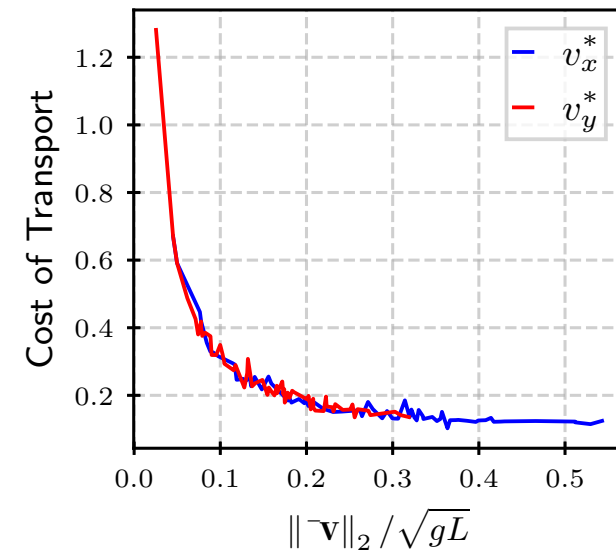


图4: 运输成本作为稳态速度范数的函数。蓝色表示沿 x 轴的速度参考值, 红色表示沿 y 轴的速度参考值。

2) 自主性标准:

a) 短期自主性: 运输成本是在机器人各种稳态速度下进行评估的。结果根据速度指令方向(向前/向后或左/右)进行分类, 并展示在图4中。

对于沿 x 轴和 y 轴的速度参考值, 运输成本随稳态速度范数的增加而降低。这可用于未来工作中为机器人的嵌入式版本建模电池消耗。此外, 它也为未来研究中的比较建立了有用的基准测试。

b) 长期自主性: 平均地面反作用力 (GRF) 被发现与机器人的速度无关, 保持在机器人重量的 $90\% \pm 6.2\%$ 。该值低于100%的原因是存在短暂的双支撑阶段——在此期间双脚短暂分担负载。由于机器人主要以单足支撑行走, 每只脚的平均地面反作用力仍低于全身重量。这表明步态转换期间地面反作用力降低, 有助于机械寿命。此外, 它证实了训练期间施加的接触力约束在实践中得以维持。

3) 鲁棒性准则:

a) 推扰恢复: 该机器人承受了超过300次推扰, 覆盖了其基座周围的全方向范围。图5展示了机器人能够承受而不摔倒的最大冲量(以机器人重量的百分比归一化表示), 并按 45° 象限进行分类。

总体而言, 该机器人对来自背部和侧面的推力表现出更强的鲁棒性。相比之下, 当从正面或角落受到推力时, 其鲁棒性通常较弱, 但右上角除外。这一异常现象可归因于该区域记录的数据点不足。作为对比, 施加相当于机器人重量4%的冲量持续一秒钟,

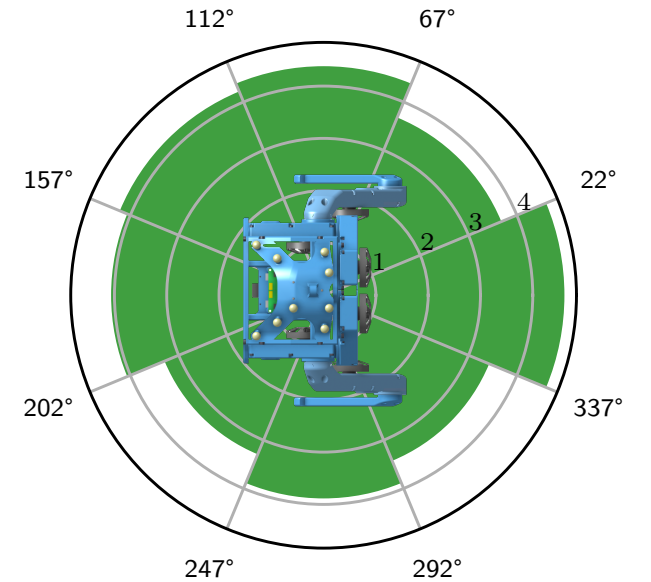


图5: 作为推扰角度的函数, 摔倒前的最大冲量(以重量百分比表示, 持续 $\times$ 秒), 其中 0° 对应于施加在机器人基座背部的推扰。

相当于一个70公斤的人在一秒钟内被一个2.8公斤的球击中, 这证明了我们方法的鲁棒性。

D. 附加结果

为了定性展示平衡过程的鲁棒性, 我们提供了附加结果。图6显示了机器人在地毯上行走时, 地毯被抽走后从滑倒中恢复的过程。这些图像直接取自随附的视频, 该视频提供了整个恢复序列更动态的可视化效果。

图7展示了机器人执行跳跃动作, 这是通过修改步态约束以允许弹道阶段实现的(见第IV-D节)。它成功完成了超过50次连续跳跃, 并保持该行为超过15秒, 如随附视频所示。

VI. 结论

本文探索了点足双足机器人的控制问题, 强调了其在实现拟人化运动中的重要性。与平足双足机器人不同, 点足机器人面临独特的稳定性挑战, 这使得其控制成为一个有趣的研究课题。

我们开发了一种约束强化学习方法, 并成功将策略从仿真迁移到真实机器人, 无需额外微调。实验表明, 该机器人能够保持平衡、跟踪速度指令、跳跃以及抵抗外部干扰, 展现出高度的鲁棒性。

尽管取得了这些成就, 但由于硬件脆弱性以及外部因素(例如使用连接线将机器人连接到外部计算机), 实际部署仍然具有挑战性。

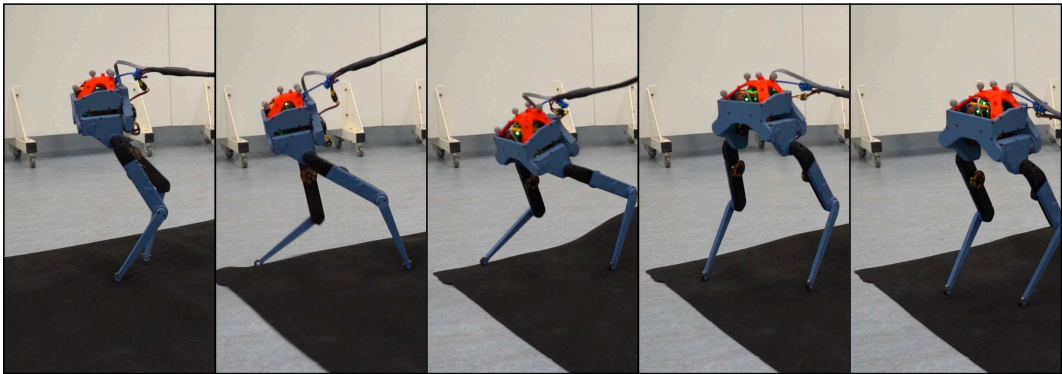


Fig. 6: Motion capture sequence of the robot recovering from a slip after the carpet was pulled away. Captured at 20 Hz.

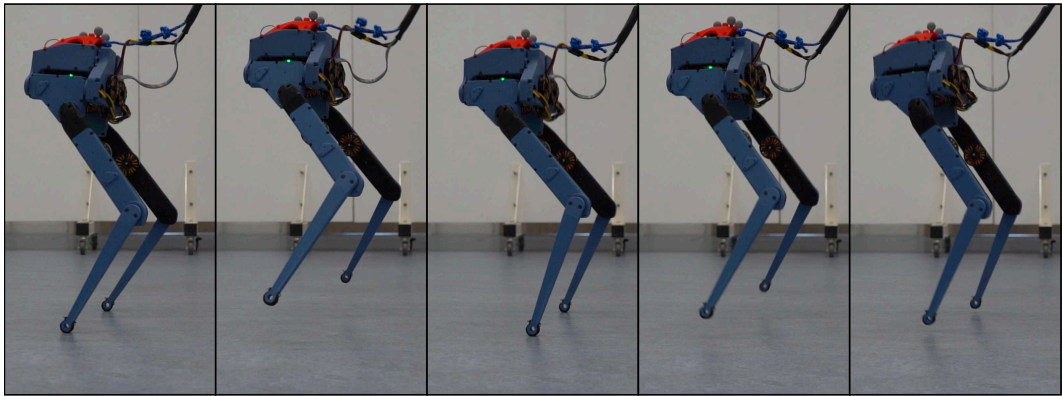


Fig. 7: Motion capture of the robot jumping. Captured at 33 Hz.

to an external computer. This setup, while necessary for communication and data processing, limits mobility and introduces additional complexities during operation. Future work will focus on refining robustness, exploring additional locomotion behaviors, and further improving real-world navigation capabilities.

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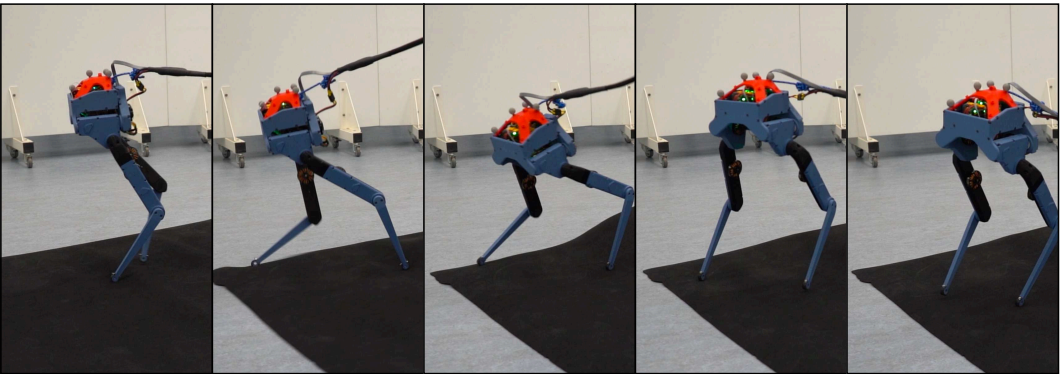


图6：机器人在地毯被抽走后从滑倒中恢复的动作捕捉序列。以20 Hz频率采集。

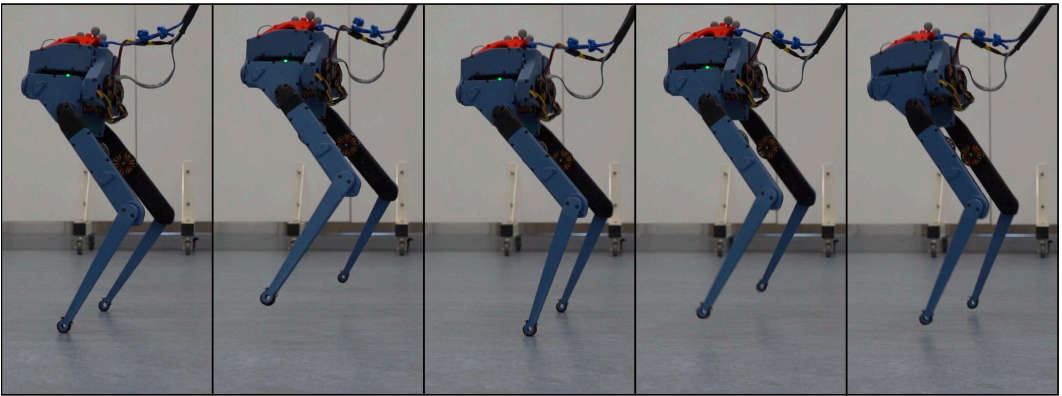


图7：机器人跳跃的动作捕捉。以33 Hz频率采集。

这种设置虽然对通信和数据处理是必要的，但限制了移动性，并在操作过程中引入了额外的复杂性。未来工作将专注于提升鲁棒性、探索更多运动行为，并进一步改进实际导航能力。

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